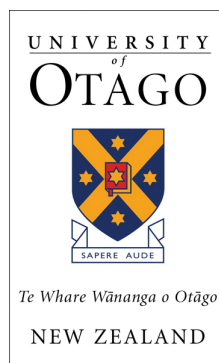


Exploring GPS Signal Data and Position Estimation Through Bayesian Inference

Tom Holmes

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Abstract

A new approach to GPS positioning has been developed wherein post processing of ultra-short sequences of captured GPS signal data can produce an estimate of receiver location. The new fixing scheme, labelled ‘FastFix’ works by downloading GPS satellite information (ephemeris) data from the internet rather than decoding the same data (at a rate of 50bps) from each satellite’s broadcast signal. Fewer than five milliseconds of signal data sampled at around 8MHz is processed in a similar manner to that which occurs in a traditional GPS receiver chip (C/A code phase and carrier Doppler shift are found through standard signal processing techniques) before a least squares optimization is used to estimate receiver position and GPS time. This thesis builds on the FastFix scheme by analyzing the binary signal data (output from the GPS receiver chip) using Bayesian inference with Markov chain Monte Carlo sampling. The outlined approach borrows ideas from the FastFix scheme to reach final solutions, giving positions typically within 250 meters of true receiver location. The Bayesian approach allows not only position and time estimates but also their relative uncertainty to be determined.

Contents

Abstract	i
List of Figures	iv
1 Introduction	1
1.1 GPS Navigation	2
1.2 GPS Signals	2
1.3 FastFix GPS	3
2 Bayesian Inference	7
2.1 The Forward Map	8
2.2 Exploring the Posterior	11
3 Searching for a Solution	14
3.1 The Doppler Fix	14
3.2 The Phase Fix	21
3.3 The Raw-data Fix	26
4 Results	31
4.1 Doppler Fix Results	31
4.2 Phase Fix Results	35
4.3 Raw-data Fix Results	36
4.3.1 Data requirements for the Raw-data fix	36
4.3.2 Raw-data fix results using 0.5ms	37

5 Discussion	40
5.0.1 Future Work	40
5.0.2 Limitations	41
5.0.3 Conclusion	42
Appendix: WGS84 to ECEF Coordinate Transformation	43
Bibliography	46

List of Figures

1.1	GPS satellite signal modulation scheme	3
1.2	Satellite Doppler shifts sum of squares over the surface of the Earth . .	5
1.3	Satellite C/A code phase sum of squares over the surface of the Earth .	6
2.1	Example of Markov chain sampling of the posterior distribution	12
3.1	Satellite carrier Doppler shifts	17
3.2	Simulated satellite carrier Doppler shifts	18
3.3	Doppler likelihood distribution of latitude and longitude parameters . .	19
3.4	Markov chain sampling of the Doppler likelihood	20
3.5	Satellite C/A code phases	23
3.6	Phase likelihood distribution of latitude and longitude parameters . . .	24
3.7	Markov chain sampling of the Phase likelihood	25
3.8	Raw-data likelihood distribution of latitude and longitude parameters .	26
3.9	Raw-data likelihood distribution with lowered sigma value	28
3.10	Markov chain sampling of the Raw-data likelihood space	30
4.1	Map overlay of Doppler fix position estimates	32
4.2	Errors and standard deviations in latitude and longitude for Doppler fixes	33
4.3	Doppler fix standard deviations vs. absolute errors	34
4.4	Map overlay of Phase fix position estimates	35
4.5	Phase fix standard deviations vs. absolute errors	36
4.6	Raw-data fix standard deviations vs. absolute errors	38
4.7	Map overlay of Raw-data fix position estimates	39

Chapter 1

Introduction

An integral part of practising science is not just finding solutions but being aware of the uncertainty surrounding them. Through the application of statistical inference for a more informative solution, this thesis is an attempt to improve upon the current methods of position determination in one application of global navigation systems. The Global Positioning System (GPS) is just one of many global navigation satellite systems in use today and has become an indispensable tool in countless areas of modern society. One increasingly popular application of GPS is the tracking or recording of animal movements. In order to create stand alone tracking devices light enough to be permanently attached to birds and other smaller mammals, the Electronics Research Group of the University of Otago has developed a novel GPS fixing scheme coined ‘FastFix’ [1] which drastically reduces a receiver’s power demands and thus necessary battery weight. One limitation of the FastFix scheme is that position information can only be calculated upon retrieval of the tracking tag. This also presents an upside, the raw signal data stored on the tracking tag can be processed (on high powered computers) to any extent desired by the user in order to obtain the position fixes. The opportunity for post-processing is ideal for the application of Bayesian inference with Markov chain Monte Carlo (MCMC) sampling, which can extract valuable information from very little data but often at the cost of heavy compute times. This thesis outlines how one may apply such techniques to retrieve not only estimates of receiver position and time, but also a measure of uncertainty in each estimated parameter. We begin

with an introduction to GPS and how FastFix works. The Bayesian approach is then introduced in Chapter 2 followed by its application to the specific problem in Chapter 3. Chapter 4 gives an account of the success of the methods employed before finally, Chapter 5 addresses the project's limitations, future work and other points of interest.

1.1 GPS Navigation

The Global Positioning System (GPS) consists of a constellation of satellites or 'space vehicles' (SV's) in non-geosynchronous orbit around the earth [2]. If at least four SV's are visible to a GPS receiver, the receiver's geographical location and synchronous GPS time may be calculated. The GPS Interface Control Document [3] contains the entire GPS signal structure specification and was used extensively throughout the work documented in this thesis. The number of satellites in operation often changes, at the time of writing we consider there to be 32 potentially active SV's.

1.2 GPS Signals

The content of this document is concerned only with single frequency GPS receivers which are designed to receive the L1 carrier signal ($f_{L1} = 1575.42\text{MHz}$). Modulated on top of each SV's carrier are two binary sequences, as shown in Figure 1.1. The first is a pseudo-random 'Gold' or 'Coarse Acquisition' code (C/A code). Each satellite's C/A code is unique and chosen such that it correlates poorly with every other C/A code [4]. This enables a receiver to identify a specific SV by correlating the received signal with a locally generated version of that satellite's C/A code. These codes are modulated onto a satellite's carrier signal using binary phase shift keying (BPSK) in order to spread the carrier's spectrum. This allows a receiver to recover a much diminished signal from below the local thermal noise floor and also allows the satellites to share the same frequency band. The 1023 bit long C/A codes repeat every millisecond and each 1ms duration is referred to as an epoch. The second binary sequence created in each satellite is the Navigation Message or data code. This contains encoded information such as the ephemeris data which includes the date and GPS time, the transmitting satellite's

position, speed and health and the location and status of every other satellite. The data code has a data rate of 50 bps, thus each data bit contains 20 C/A code epochs. The C/A code correlations performed by the receiver give a measure of each visible satellite's C/A code phase (represented as a fraction of an epoch). These phases are used to assist the receiver in 'locking on' to each satellite's data code using a phase locked loop (PLL). The data codes can then be extracted from the signal and the satellite's ephemerides are stored. In order to obtain a receiver position and time fix, a pseudorange is calculated for each visible SV by multiplying their signal's time of flight (the delay from the time of transmission by satellite to reception at receiver antenna) by the speed of light. The 'pseudo' refers to the error in calculated satellite range caused by the time of reception, as measured by a receiver's onboard real time clock, being not yet synchronized with GPS time. This timing error is the reason four GPS satellites are required to produce an accurate position estimate, one for each of the unknown parameters, latitude, longitude, altitude and time.

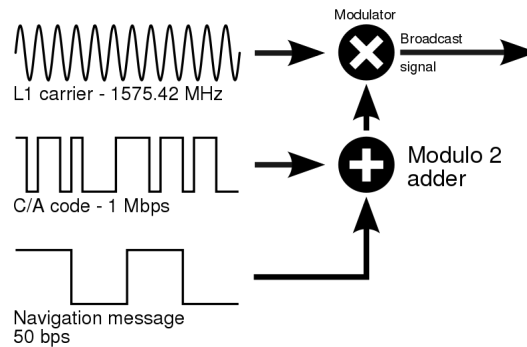


Figure 1.1: GPS satellite signal modulation scheme. Each satellite's C/A code and navigation message are modulo-2 added before being multiplied with a 1.575GHz carrier signal.

1.3 FastFix GPS

With no prior information, a GPS receiver requires as much as 12.5 minutes (the repetition time of each Navigation Message) of signal data to calculate an initial position fix.

The size and weight of a receiver's power source (most commonly batteries) required to meet the demands of this amount of signal reception and processing, constrain the lower limit of receiver's overall size and weight. In GPS applications where receiver miniaturization is important, such as tracking devices for birds or small mammals, an alternative fixing scheme has been developed. By downloading the broadcast ephemerides data (as contained in each Navigation Message) from the internet, rather than decoding it from a satellite's signal, 'FastFix GPS' can obtain a fix using as little as a few milliseconds of recorded signal data [1]. By recording only four milliseconds of signal data every five minutes (or as often as a fix is required), a receiver's power demands are dramatically reduced. This allows the use of a much smaller battery but also requires that the receiver be physically recovered and its stored data post-processed to calculate the fixes.

The first stage of this data processing is referred to as the Doppler fix. As each of the GPS satellites traverse their unique elliptical orbits, their velocity vectors continually change. If one draws a line between a GPS receiver on earth and a particular SV, the component of that SV's velocity vector along this line determines a shift in the satellite's emitted signal frequency (Doppler shift) as seen by the receiver. From the few milliseconds of FastFix data, the magnitudes of each visible SV's Doppler shift can be found. From the downloaded ephemerides, one can find a satellite's position and velocity at any given time. This can then be used to calculate each satellite's *expected* Doppler shift for any given receiver location. By minimization of the sum of squared differences (often termed 'least squares') between expected and measured Doppler shifts for each proposed receiver position, one can estimate a position fix to one standard deviation of around 100km (using 4ms of captured signal sampled at 8.184MHz). For the 4ms of data recorded at 7.22pm on 19th May 2010, Figure 1.2 shows one such sum of squares space over the surface of the earth.

Similar to the least squares Doppler fix, the next stage of the FastFix scheme is the Phase fix. The phase of each SV's C/A code can be extracted from the FastFix data (through correlation with locally generated C/A codes) and compared to expected code phases (calculated using the downloaded ephemerides) to give a more precise

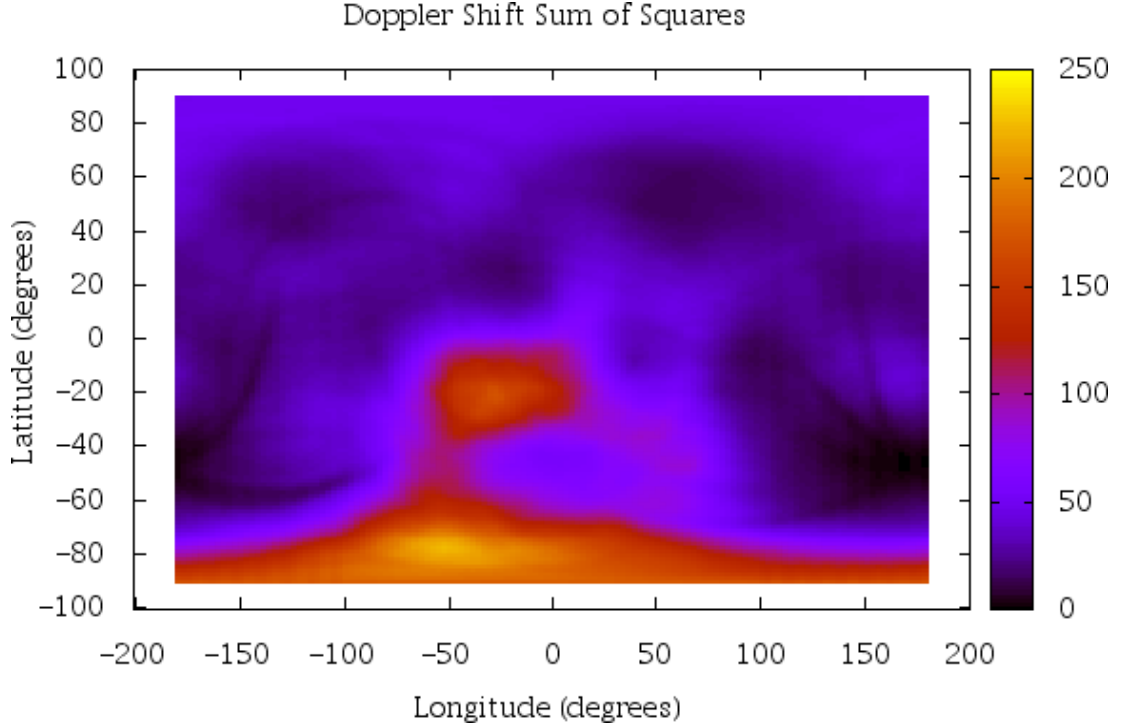


Figure 1.2: Sum of squared differences between expected and measured satellite Doppler shift values over the surface of the earth. Darkest region around latitude -46 and longitude 170 indicates the GPS receiver is most likely to be somewhere in the South Island of New Zealand.

fix, often to a standard deviation of around 1km. Since each C/A code is repeated every millisecond, we see periodic troughs in the C/A code phase sum of squares space separated by 300km ($= 3 \times 10^8 \text{ms}^{-1} \cdot 1 \times 10^{-3} \text{s}$). This leads to a structured multimodal space as shown in Figure 1.3. The axes of Figure 1.3 are relative to the receiver's true position, upon the roof of the Science III building, University of Otago, New Zealand (Latitude: -45.8639, Longitude: 170.5133).

As part of the FastFix régime, a Nelder-Mead (downhill simplex) method is implemented for the above two minimization problems. Since the Nelder-Mead method is best suited to unimodal distributions, it reliably provides a position estimate of the minimum of the Doppler shift sum of squares space. However, due to the multimodal nature of the phase fix space, a Doppler fix must be within the immediate vicinity of

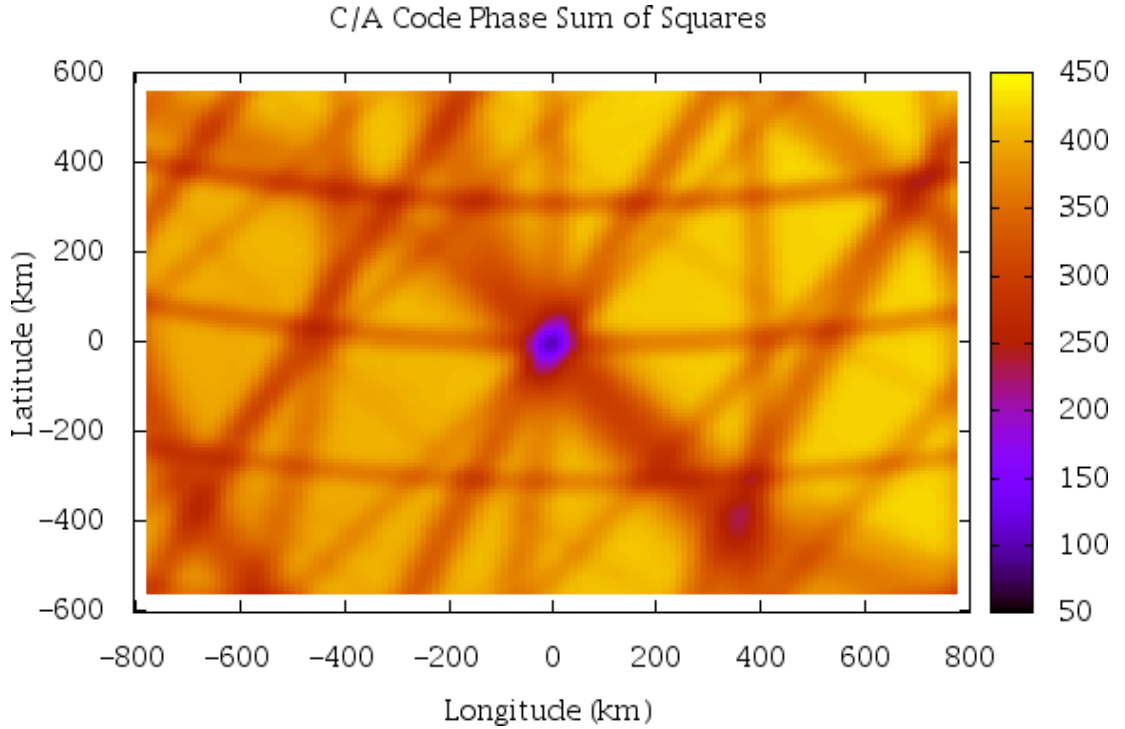


Figure 1.3: Sum of squared differences between expected and measured satellite C/A code phases in the vicinity of the true receiver location (dark spot in center). Latitude and longitude axes are relative to the receiver position. The dark red ridges are caused by the periodicity of the C/A codes and can be thought of as representing wavefronts from each satellite.

the correct C/A code phase sum of squares minimum to provide an accurate solution. Given that a least squares minimization provides only an estimate of the minimum, that is, no information regarding the local structure of the search space is obtained, the accuracy of a specific Doppler fix is unknown. If a poor Doppler fix were to lead to an incorrect Phase fix, there is no way to identify such an event and have it flagged as an outlier. This is a major drawback of the FastFix process and is one of the main incentives to investigate alternative data processing methods, such as statistical inference.

Chapter 2

Bayesian Inference

The main objective of this thesis is to approach the problem of GPS fix acquisition armed with more than signal processing techniques and trilateration. Inverse problem theory involves using observations of some phenomena to glean information on possible causes [5]. For the case of GPS fixing we create a model for the physical system which we call the forward map $F_k : \mathbf{x} \mapsto y_k$, as it maps the model parameters $\mathbf{x}$ to the N data samples y_k , $k = 1, 2, \dots, N$. The model parameters of most interest to us are the GPS receiver's position and time, the observations being the binary data at the receiver output. As our data is subject to random variation (the amplitudes of the signals of interest are well below the thermal noise floor), this falls into the category of statistical inference [6]. One way to tackle a problem in statistical inference is to employ Bayes' Theorem. Expressed in terms of probability density functions for continuous parameters,

$$f(\mathbf{x}|y_k) = \frac{1}{f(y_k)} f(y_k|\mathbf{x}) f(\mathbf{x}) \quad (2.1)$$

where $f(\mathbf{x})$ is the prior pdf (probability density function), $f(y_k|\mathbf{x})$ is the likelihood pdf and $f(\mathbf{x}|y_k)$ the posterior pdf. The $\frac{1}{f(y_k)}$ term can be thought of as a normalization factor and is often left out, giving instead

$$f(\mathbf{x}|y_k) \propto f(y_k|\mathbf{x}) f(\mathbf{x}). \quad (2.2)$$

The prior probability $f(\mathbf{x})$ represents everything we know about our model parameters $\mathbf{x}$ before the data y_k has been collected. For example, if the GPS receiver were in

the form of an animal tracking tag, we can be fairly certain that the receiver's altitude will be non-negative, as GPS receivers don't tend to function well underwater. Also we can restrict the altitude to less than a few thousand feet depending on the nature of the animal being tracked. The accuracy of a real time clock on board the receiver can also be expressed in the form of a probability density function and is a vital piece of prior knowledge. Evaluation of the forward map $F_k(\mathbf{x})$ for a given state of the parameters $\mathbf{x}$ gives us the noise free data $\bar{y}_k$ we would expect to see, provided the forward map correctly modelled the system. Assuming a basic noise model of additive Gaussian noise with standard deviation σ , ie. $y_k = \bar{y}_k + \mathcal{N}(0, \sigma)$, we find the following expression for the likelihood,

$$f(y_k|\mathbf{x}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2} \sum_{k=1}^N [y_k - F_k(\mathbf{x})]^2\right). \quad (2.3)$$

Multiplying the likelihood by the prior gives (up to a normalization constant) the posterior distribution $f(\mathbf{x}|y_k)$, which tells us the probability that the true values of $\mathbf{x}$ lie within a certain range. Our ultimate goal is to calculate statistics which describe the posterior distribution in order to estimate $\mathbf{x}$.

2.1 The Forward Map

Our forward map begins with four parameters, the receiver's latitude, longitude and altitude plus GPS time. Combining these with the satellite ephemerides data, we can calculate each SV's expected C/A code phase, Doppler shift and signal amplitude. The first step required is to convert the receiver's position from WGS84 coordinates (latitude, longitude and altitude on an ellipsoid) into the ECEF coordinate system (earth centered, earth fixed) as used in the ephemerides to describe the GPS satellite positions and velocities. This is a well documented and understood conversion which has been summarized in the Appendix. The signal emitted from each satellite begins as a sine wave carrier multiplied by the satellite's C/A code, $C(t)$, giving

$$aC(t_t) \sin(\omega_c t_t) \quad (2.4)$$

where ω_c is the angular frequency of the L1 carrier signal and a is the transmitted

signal amplitude. Note that the satellite's data code has been left out as it only has possible bit transitions every 20 milliseconds. This introduces the possibility for some C/A codes to appear inverted and is dealt with in another part of the final code. The time of transmission, t_t , is equal to the time of reception, t , minus the time of flight, τ , giving

$$aC(t - \tau) \sin(\omega_c t - \omega_c \tau). \quad (2.5)$$

A signal's time of flight can be broken down in two ways: into an integer number of C/A code epochs plus a remainder portion, or as an integer number of carrier frequency cycles plus a remainder. Expressing this mathematically,

$$\tau = (n + \delta_{C/A})T_{C/A} = (m + \delta_{L1})\frac{2\pi}{\omega_c} \quad (2.6)$$

where $T_{C/A}$ is the C/A code period (1ms), $n, m \in \mathbb{N}$ and $\delta_{C/A}, \delta_{L1} \in [0, 1)$.

At this point it is convenient to split the absolute time parameter t into two parts. The first, which we call t_z as it will always begin at zero, is the reception time relative to the closest C/A code epoch beginning. The second, the C/A code time offset, is simply the difference between the reception time and the closest C/A code epoch beginning. The C/A code time offset will be expressed as $\delta_t T_{C/A}$ where $\delta_t \in [0, 1)$, as it represents a fraction of one C/A code epoch. All SV's clocks are synchronized (with atomic clock precision) such that each will begin transmission of every CA code epoch at the same instant, thus $\delta_t T_{C/A}$ is common to all SVs. Similarly, inside the sine function we split t into the same t_z and a constant phase offset ϕ . As the Doppler shifts for each SV signal are different and ϕ is relative to the Doppler shifted carrier frequency, this introduces 32 new phase offset parameters into the forward map (one carrier phase offset for each SV). The introduction of new parameters may seem counter-intuitive as it increases model complexity, but as will be shown later, it serves to 'smooth out' the likelihood space by removing the C/A code phase and signal carrier phase perturbations. It is worth noting that the absolute time still plays a role in the forward map, as it determines the positions of the satellites and thus their expected C/A code phases and Doppler shifts. Combining 2.5 and 2.6 we get

$$aC(t_z + [\delta_t - n + \delta_{C/A}]T_{C/A}) \sin(\omega_c t_z + \phi - \omega_c(m + \delta_{L1})\frac{2\pi}{\omega_c}). \quad (2.7)$$

Since both $C(t)$ and $\sin(t)$ are periodic, with periods $T_{C/A}$ and $\frac{2\pi}{\omega_c}$ respectively, the integers n and m may be removed. Everything in the sine function with no dependence on t_z can be absorbed into a phase constant, ϕ , simplifying 2.7 down to

$$aC(t_z + [\delta_t - \delta_{C/A}]T_{C/A}) \sin(\omega_c t_z + \phi). \quad (2.8)$$

For simplicity we ignore the effects of the Earth's ionosphere on the propagating signal (a small time delay and frequency shift), and assume a basic model of signal amplitude $A(\theta)$ (absorbing a from above) which depends only on the satellite's angle of elevation, θ . Adding in the signal's Doppler shift ω_{ds} , we get at the receiver's antenna,

$$A(\theta)C(t_z + [\delta_t - \delta_{C/A}]T_{C/A}) \sin(t_z[\omega_c + \omega_{ds}] + \phi). \quad (2.9)$$

Multiplication of the signal with a local oscillator inside the receiver then down-samples the L1 carrier to a more manageable near DC frequency which we will call the center frequency ω_{cf} . This is the frequency at which we would see a carrier signal that had no Doppler shift. The signal is filtered to retain only the carrier's bandwidth (about 2Mhz), then I and Q sampled at a user configurable sampling frequency ω_s , giving

$$A(\theta)C(t_{z_k} + [\delta_t - \delta_{C/A}]T_{C/A}) \exp(jt_{z_k}[\omega_{cf} + \omega_{ds}] + \phi). \quad (2.10)$$

The sampling process involves digitizing to one bit precision, thus combining the signals from all satellites m , we get

$$H \left(\sum_{m=1}^{32} \left[A(\theta^m)C(t_{z_k} + [\delta_t - \delta_{C/A}^m]T_{C/A}) \exp(jt_{z_k}[\omega_{cf} + \omega_{ds}^m] + \phi^m) \right] \right) \quad (2.11)$$

where $H(x)$ is the Heaviside step function,

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

and the sampling frequency ω_s determines the spacing between samples, ie.

$$t_{z_k} - t_{z_{k-1}} = \frac{2\pi}{\omega_s}.$$

The forward map now requires as inputs more than just the three position parameters and time. Parameters in which we have no interest are often termed nuisance

parameters. These include the receiver center frequency ω_{cf} , sampling frequency ω_s , C/A code phase offset δ_t and the 32 signal carrier phases ϕ^m , giving a total of 39 dimensions in our state space.

2.2 Exploring the Posterior

A common approach for exploring complex probability distributions is to use Markov Chain Monte Carlo (MCMC) sampling. Two major advantages of MCMC methods are their simplicity and adaptability. See [7] for a great introduction and reference text. The Metropolis Hastings algorithm was used with a random walk to sample our posterior distribution. The random walk MCMC, as the name suggests, takes steps of random size and direction in each dimension of the target distribution. Each proposed step is either accepted or rejected, ie. the chain changes or stays the same, with a probability of acceptance proportional to the extent that the step leads to a point of higher probability density, ie. the chain is more likely to move ‘uphill’. In this way, our Markov chain spends more time in regions of higher probability density than it does lower. Plotting the chain’s parameter states at each step in histograms gives an approximation of each parameter’s marginal distribution. Figure 2.1 shows a typical output after running an MCMC on the posterior distribution.

The random walk MCMC method used is most simply applied in situations where the target distribution is uni-modal and decays smoothly away from the mode. This allows the chain to follow a positive gradient to find the mode. If, instead, the distribution decays rapidly from the mode, ie. a sharp peak, then has a near zero gradient for the remainder of the search space, the chain has difficulty being steered toward the mode and may never reach it. Searching the surface of the Earth for a Gaussian shaped peak with a standard deviation of around 30 meters can be time consuming. Include 37 more dimensions into the search space and we truly have a ‘needle in a haystack’ situation. This indeed posed a problem for sampling our posterior distribution, with the chain, more often than not, failing to finding the correct mode even after millions of steps. To tackle such a problem we can use more informative priors, which serves to reduce the search space, or perform some reduction on the data. By using less data

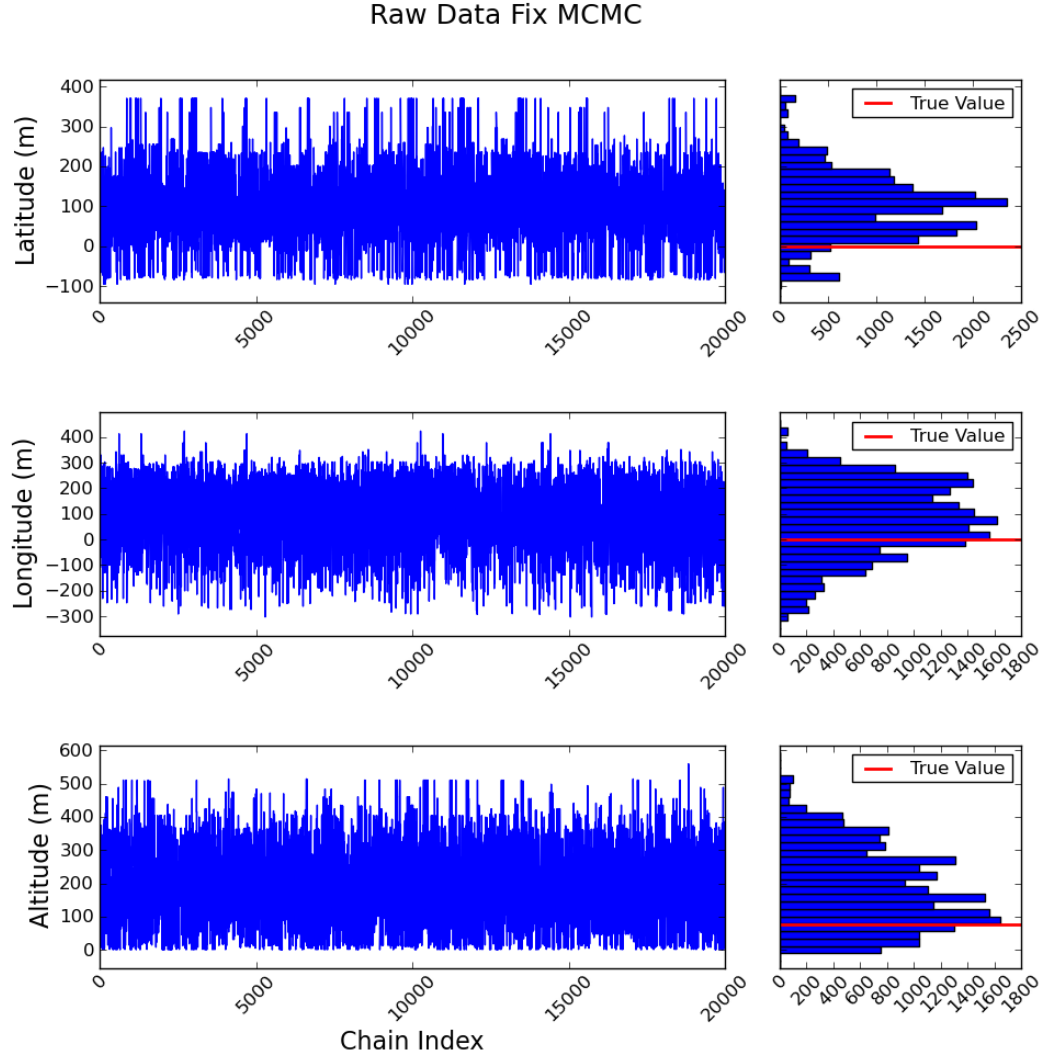


Figure 2.1: Typical Markov chain of length 20000 (excluding burn-in period) exploring the posterior distribution. Histograms on right approximate each parameters marginal distribution where the latitude and longitude parameters are relative to their known true values. States and histograms of nuisance parameters not shown.

we effectively widen the peak for which we are searching. As explained in the following section, a combination of these two concepts was used.

Chapter 3

Searching for a Solution

3.1 The Doppler Fix

In order to reliably calculate a position fix, the inverse problem was broken down into three stages. The first step we call the Doppler fix and is based on the process outlined in the Fastfix GPS section 1.3. The Doppler fix step is essentially a data reduction which aids in the search for the likelihood mode, as discussed at the end of the previous chapter. The raw signal data obtained from the GPS receiver is cross-correlated with each of the 32 SV C/A codes, with each correlation repeated for a range of possible Doppler shifts, covering the receiver's center frequency $\pm 6\text{kHz}$. Figure 3.1 shows the Doppler shifts for the first eight satellites using data recorded at 7.22pm on 19th May 2010. Since the height of the peaks is the maximum magnitude of the correlation, it is proportional to signal strength.

In essence we have added another step to our forward map and should rewrite it accordingly. However, correlating C/A codes for 32 SVs for 50 possible Doppler shifts each, takes significant CPU time ($\sim 20\text{s}$), making thousands of forward map evaluations impractical. To simplify matters, we can create a new approximate (Doppler) forward map F_d which takes a parameter vector $\mathbf{x}_d$, consisting of the receiver's position, GPS time and center frequency. Rather than producing a binary signal vector as noise free data output, the Doppler forward map uses only the calculated Doppler shifts and satellite elevation angles (to determine strength) to simulate data in the form of Figure

3.1. We will refer to the reduced (real) data as $\mathbf{y}_d$. The result of evaluating the Doppler forward map at the correct receiver location and time (corresponding to the data used to produce Figure 3.1) is shown in Figure 3.2. The error associated with the Doppler shift data is not simply additive Gaussian noise, however, treating it as such to obtain a Doppler likelihood function L_d , still holds merit. Our reduced data Doppler likelihood function is expressed as

$$L_d(\mathbf{y}_d|\mathbf{x}_d) \propto \exp\left(-\frac{[\mathbf{y}_d - F_d(\mathbf{x}_d)]^t[\mathbf{y}_d - F_d(\mathbf{x}_d)]}{2\sigma_d^2}\right). \quad (3.1)$$

Maximizing 3.1 is equivalent to minimizing the sum of squared differences between the real reduced data $\mathbf{y}_d$ and the simulated reduced data $F_d(\mathbf{x}_d)$ [5]. Figure 3.3 shows the structure of this Doppler likelihood in the vicinity of the receiver's true position, conditional on the parameters not plotted having been set to their estimated true values.

Although 3.1 is by no means an accurate depiction of the true forward probability density, it still holds valuable information. As the Doppler forward map is a function of five parameters (latitude, longitude, altitude, time and center frequency), using the random walk MCMC described previously is a convenient method for exploring it. Figure 3.4 shows the result of running the MCMC on the Doppler likelihood. Our prior knowledge of GPS system time is given by the real time clock onboard the receiver. Although this time often has an accuracy of only one or two seconds, it is more reliable than that provided by the Doppler fix (~ 50 s standard deviation), thus the histogram of chain states for the time parameter provides no new information and is omitted from the figure. Similarly, our prior knowledge of receiver altitude ($0 < altitude < 1000$ m for example) is more accurate than that provided by a Doppler fix (~ 50 km standard deviation). This particular Doppler fix gives a mean latitude and longitude of (-45.4948, 169.6732) which, ignoring altitude error, is 77.1km away from the true position (-45.8639, 170.5133). The standard deviation of the latitude states in the chain is 0.6716 degrees (= 74.70km) and the standard deviation of the longitude is 1.7238 degrees which at -45.4948 degrees latitude is equivalent to 134.6km. These are typical Doppler fix values, as outlined in Chapter 4.

The means and standard deviations calculated from the Doppler fix Markov chain are used to approximate Gaussian distributions for each parameter. These are passed

to the next step of the fix process, the Phase fix, as prior distributions. If the standard deviations are over a certain limit, set by the user and discussed in the Results chapter, the Doppler fix can be flagged as a poor fix and further processing can be aborted. This remedies one of the principle issues of the Fastfix approach outlined in Section 1.3.

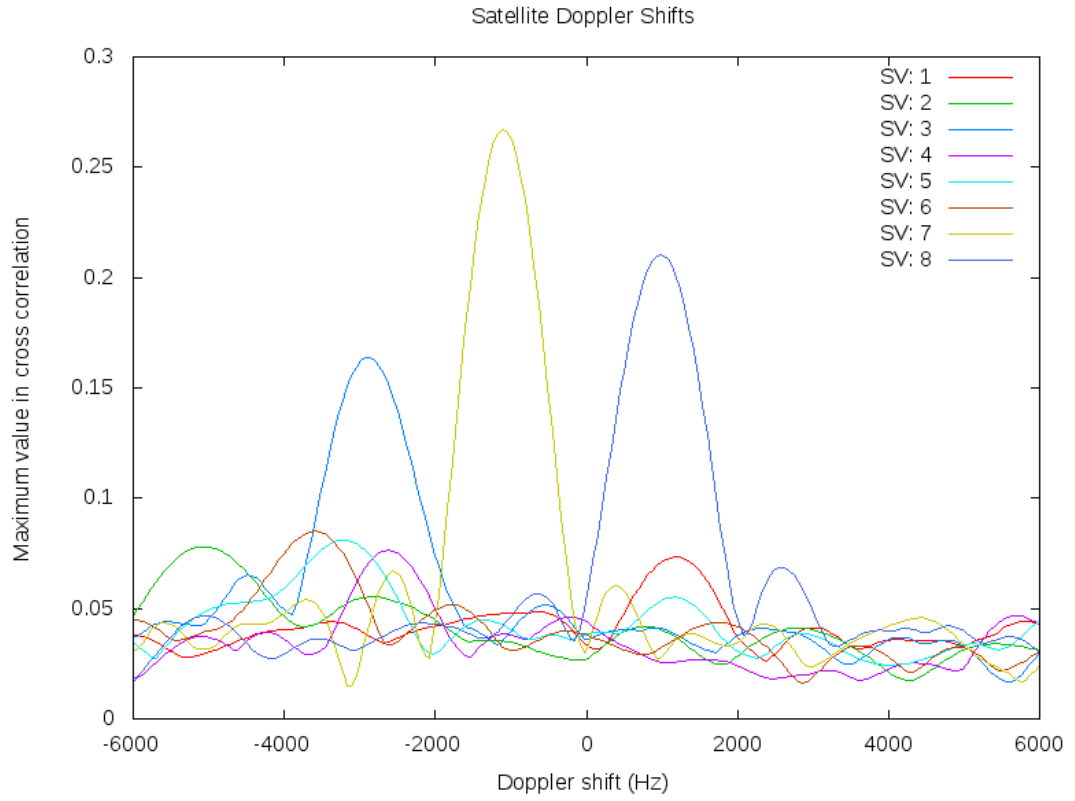


Figure 3.1: The received signal data is multiplied by a locally defined carrier signal at each possible (Doppler shifted) frequency. This removes the carrier from the signal leaving just the C/A code (plus the data message which we can ignore) which is then cross correlated with a locally defined replica. By plotting the maximum value from the C/A code correlation for each possible Doppler shift, we find the Doppler shifts for visible satellites. For example, the dominant yellow peak indicates the signal from SV number 7 is the strongest visible and has been Doppler shifted by around -1000Hz .

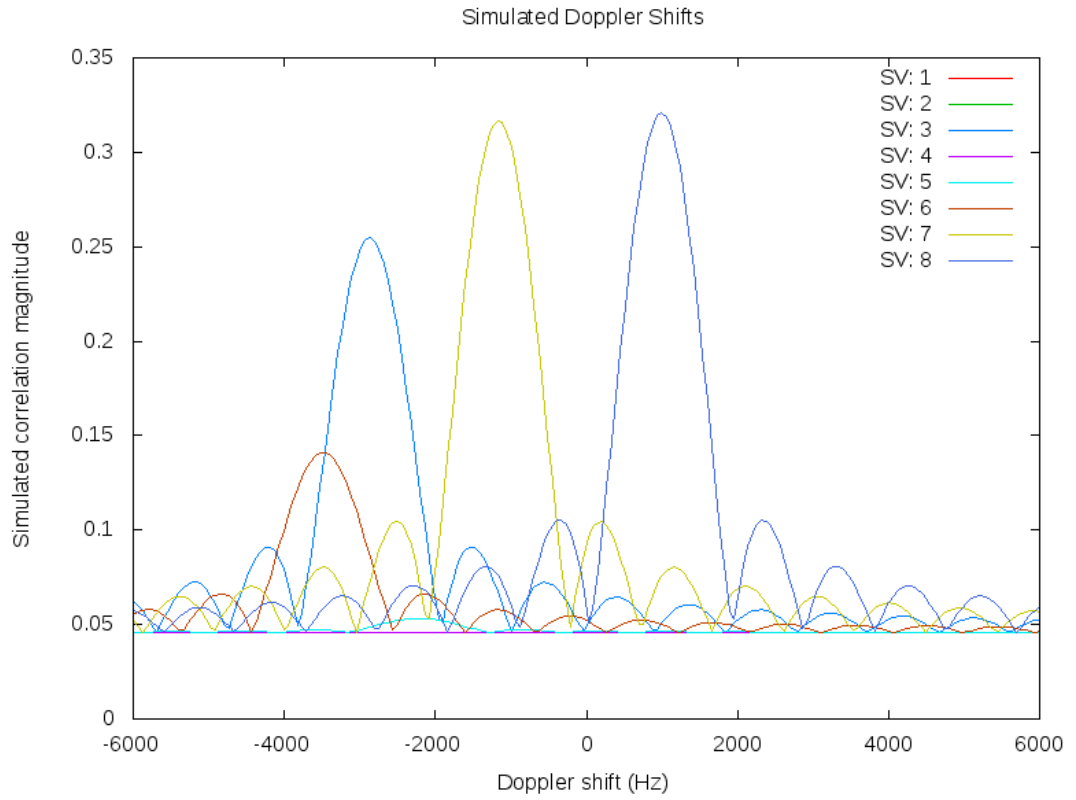


Figure 3.2: Simulated Doppler shifts produced by evaluating the Doppler forward map at the correct receiver position. By comparing with Figure 3.1 we see the Doppler forward map gives a close replica of the true data.

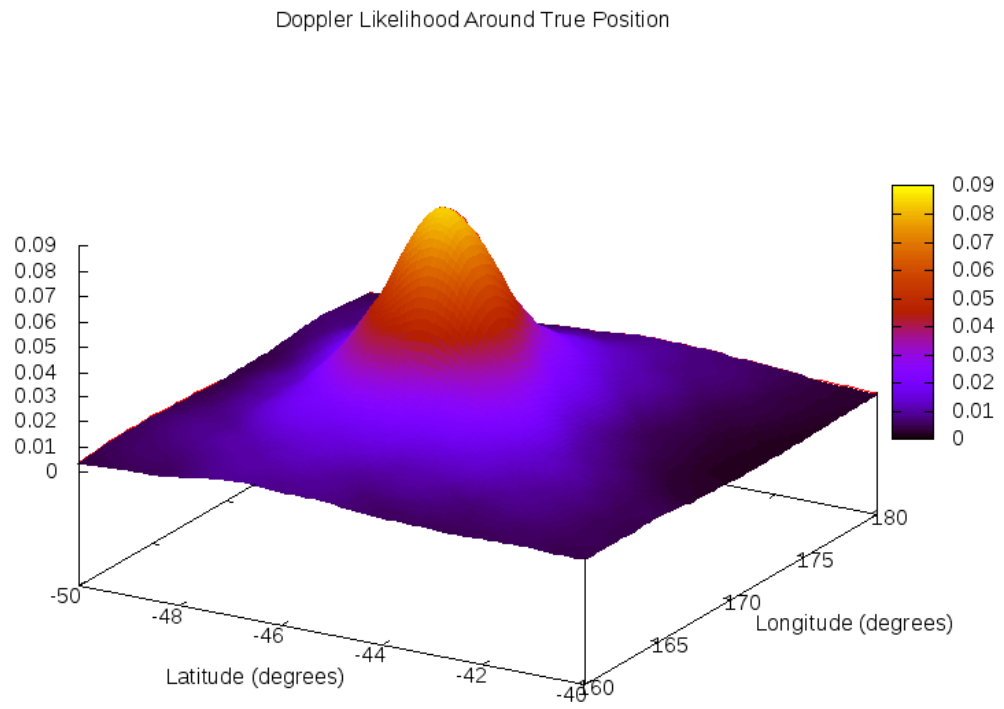


Figure 3.3: Doppler likelihood distribution of latitude and longitude parameters (conditional on set values of all other forward map parameters) with (roughly) Gaussian shaped peak centered around the true receiver location. Maximum value is 23km away from the true position.

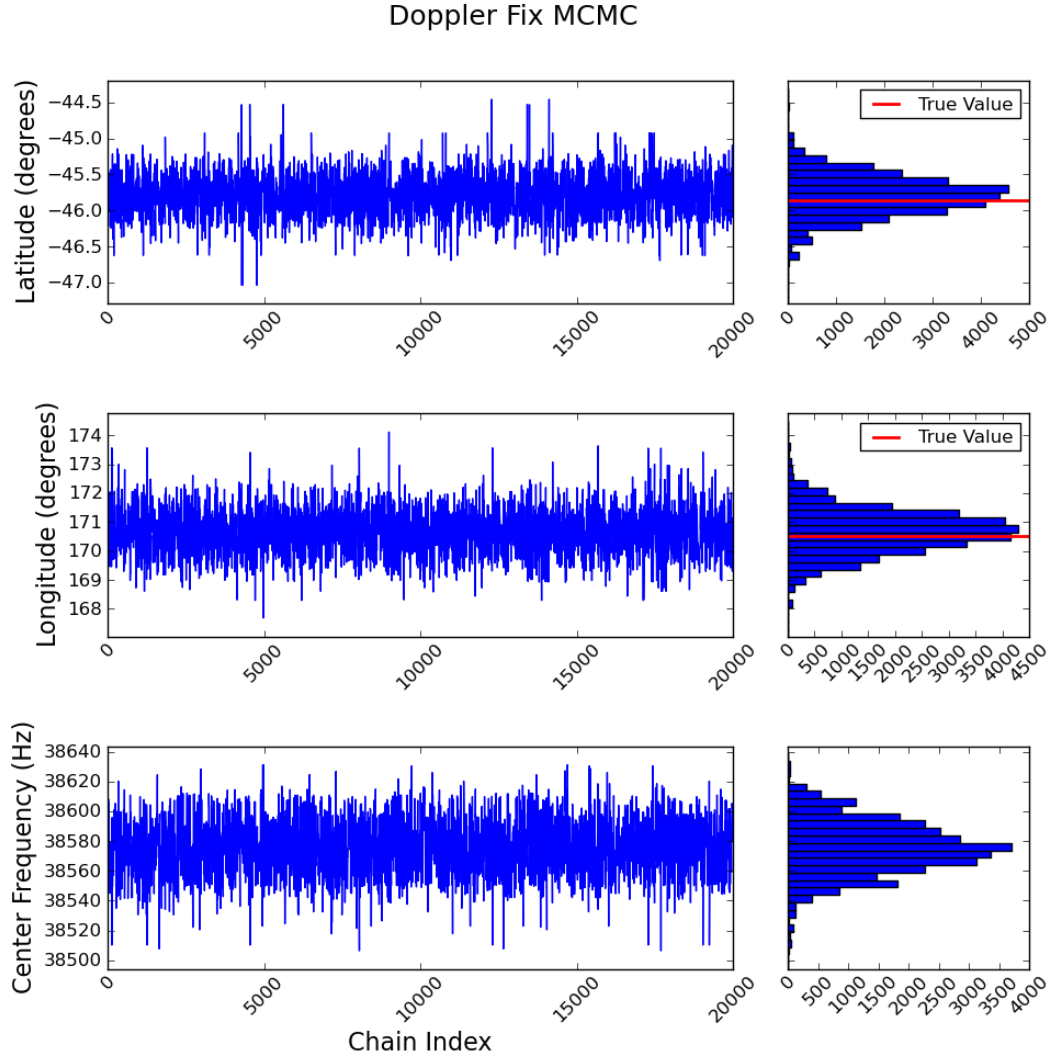


Figure 3.4: Running a Markov chain of length 20000 (excluding burn-in) on the Doppler likelihood (of Figure 3.3), we can plot histograms of the states occupied by each parameter. These histograms approximate the parameter’s marginal distributions. The histograms for the latitude and longitude states in the example can be closely approximated with Gaussian distributions, the means of which are seen to lie very close to the true parameter values (indicated by the red lines).

3.2 The Phase Fix

The Phase fix step is conceptually equivalent to the Doppler fix, a new (Phase) forward map has been created to reflect a data reduction process. First, we have some prior knowledge of parameters (the Gaussian distributions produced from the Doppler fix). The likelihood function associated with the Phase forward map is then explored using the random walk MCMC, the resulting chains are used to approximate Gaussian distributions for each parameter and finally these are passed as prior information to the next and final step, the Raw-data fix. The reduced data used in the Phase fix step, denoted $\mathbf{y}_p$, is derived from the same C/A code correlations used in the Doppler fix. In this case, however, for the Doppler shift value which produces the highest correlation magnitude, we take the entire correlation vector and use the C/A code phase information it contains. Such data for the first eight satellites is shown in Figure 3.5.

Just as in the Doppler fix, the new Phase forward map F_p is created to simulate the reduced data for a given state of the input vector $\mathbf{x}_p$. The parameters of $\mathbf{x}_p$ are the receiver's latitude, longitude, altitude and time plus the global C/A code phase offset parameter δ_t , which was introduced in the original forward map outlined in section 2.1. The likelihood function L_p for the Phase fix,

$$L_p(\mathbf{y}_p|\mathbf{x}_p) \propto \exp\left(-\frac{[\mathbf{y}_p - F_p(\mathbf{x}_p)]^t[\mathbf{y}_p - F_p(\mathbf{x}_p)]}{2\sigma_p^2}\right), \quad (3.2)$$

has structure (conditional on the set altitude, GPS time and C/A code phase offset) shown in Figure 3.6 for the example data file.

We explore this likelihood using a random walk MCMC, with Figure 3.7 showing a typical result. The fix given by the latitude and longitude means of this example (-45.8642, 170.5126) is now only 61.9 meters away from the true position. The standard deviations of the latitude and longitude are 262 meters and 359 meters respectively. Again these are typical values for the Phase fix step. On top of the latitude and longitude estimates we can now quote an estimate of altitude and GPS time which is more accurate than our prior knowledge. The mean of our GPS time offset tells us (with an associated uncertainty given by the standard deviation) how much the receiver's onboard real time clock is out by. Since the altitude's pdf (as approximated

by the histogram of the Markov chain's altitude states) is not well approximated by a Gaussian (it is closer to a Gaussian multiplied by a unit step function), the mean and standard deviation of the states of the altitude parameter in the chain make poor estimators of the true altitude and its uncertainty. Instead, a new function must be fitted to the histogram to estimate these. This would be rather straight forward but was low on the priority list of fix requirements so has not yet been implemented.

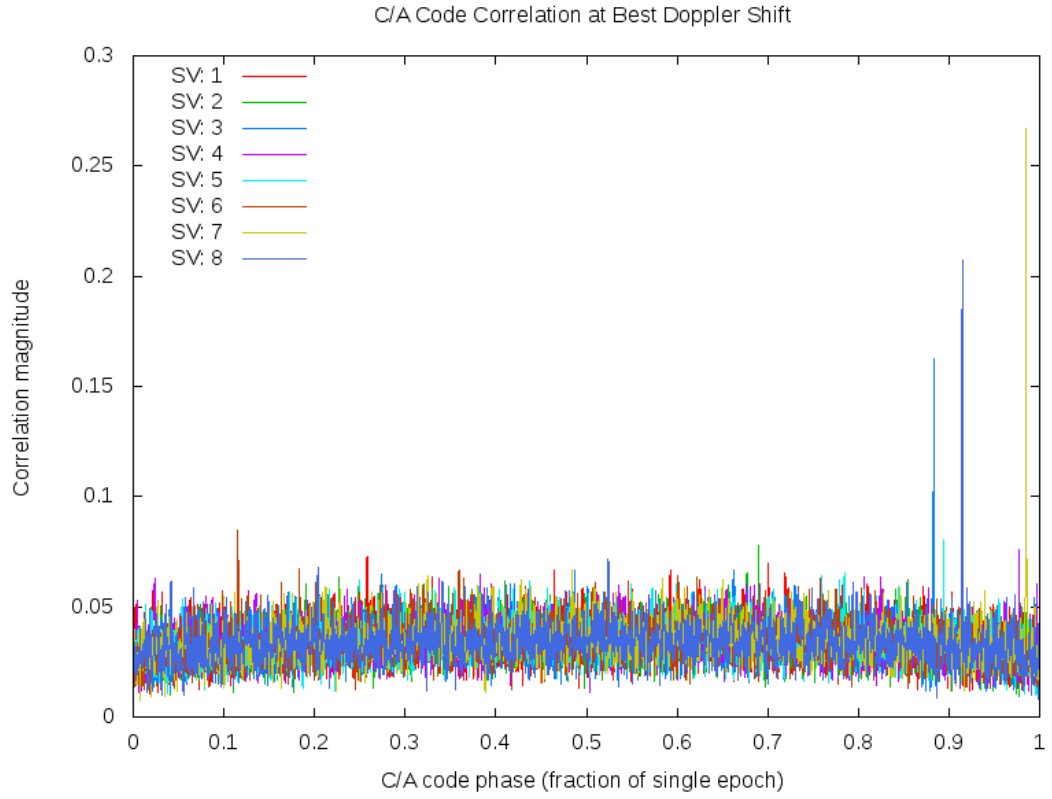


Figure 3.5: Using the same correlation data from the Doppler fix step, for each SV we take the correlation vector which had the highest maximum value (which corresponds to using the correct Doppler shift value to remove the carrier). These C/A code correlation vectors provide each SV's C/A code phase. Plotted here are the cross correlation vectors for the first 8 satellites. We see that SV 7 with the strongest signal has a C/A code phase of around 0.98.

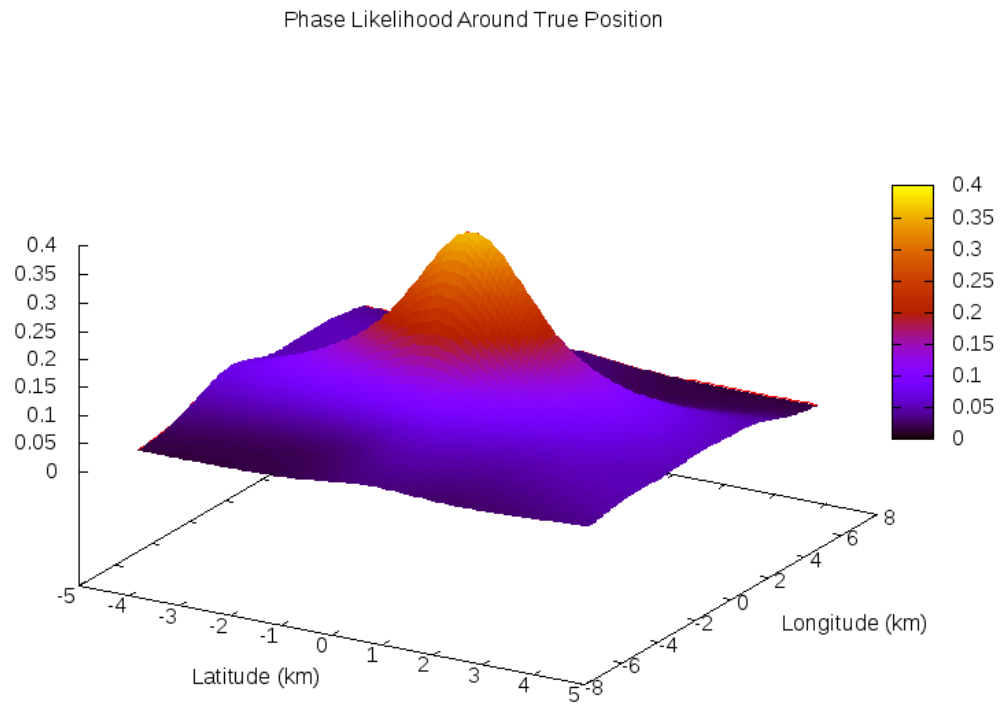


Figure 3.6: Phase likelihood distribution of latitude and longitude parameters (conditional on set values of every other forward map parameter) with (roughly) Gaussian shaped mode. The latitude and longitude axes are relative to the true receiver position. Maximum value is 217m away from the receiver location.

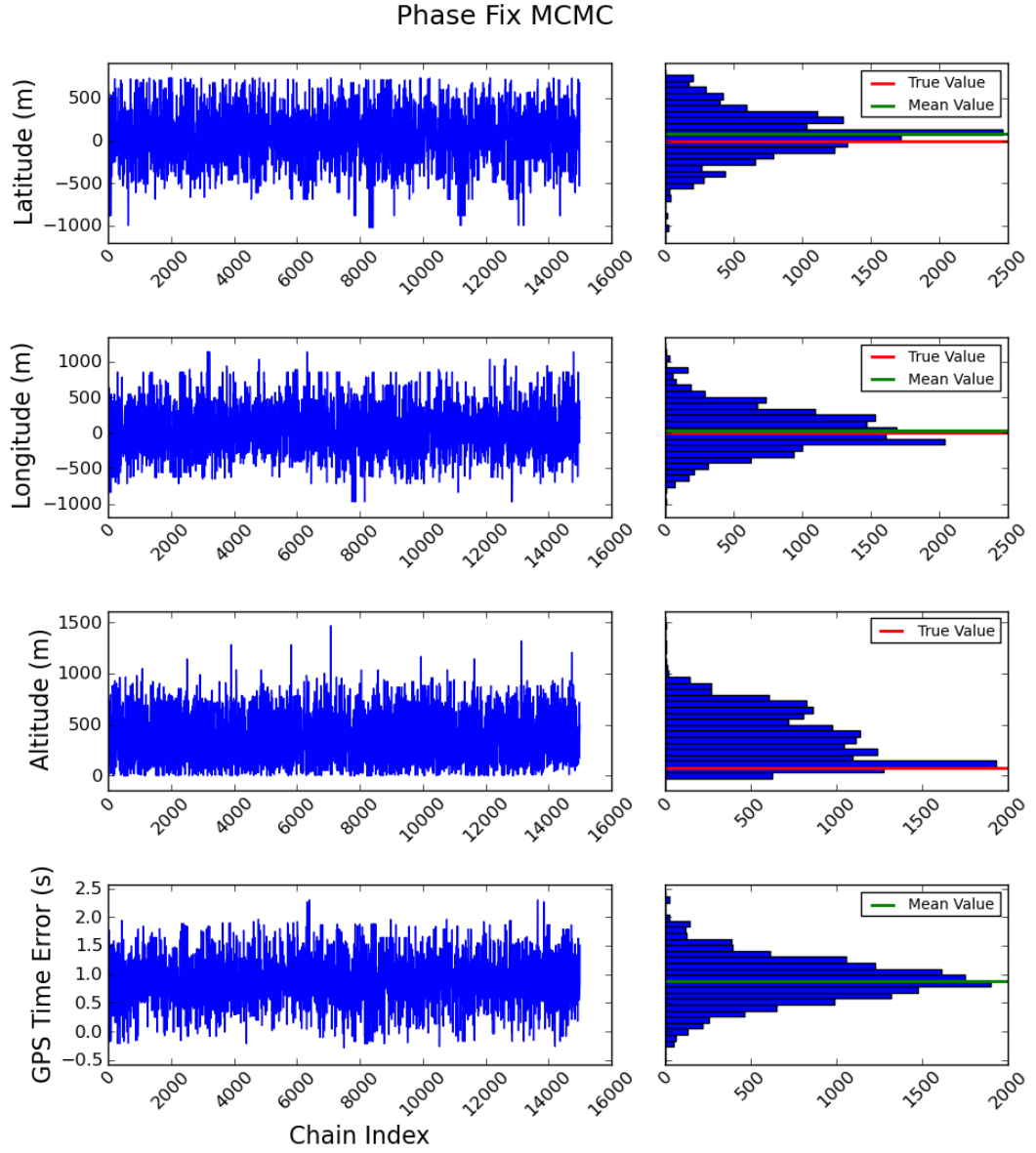


Figure 3.7: Running a Markov chain of length 15000 on the Phase likelihood. The latitude and longitude here are relative to the true value in meters. The histograms for the latitude, longitude and GPS time error states in the example can be closely approximated with Gaussian distributions. The means of these Gaussian distributions are shown as green lines along with the known true values (for those parameters we do know) as red lines.

3.3 The Raw-data Fix

The final stage of the positioning algorithm is the ‘Raw-data’ fix. Here we compare the binary I and Q data that is captured from the output of the GPS receiver chip with the data simulated from our initial system forward map from Section 2.1. Figure 3.8 shows the structure of the likelihood function around the known receiver position. The raised ridges correspond to wavefronts of signals from each satellite, therefore they should all overlap (increasing the probability density) at the true receiver position.

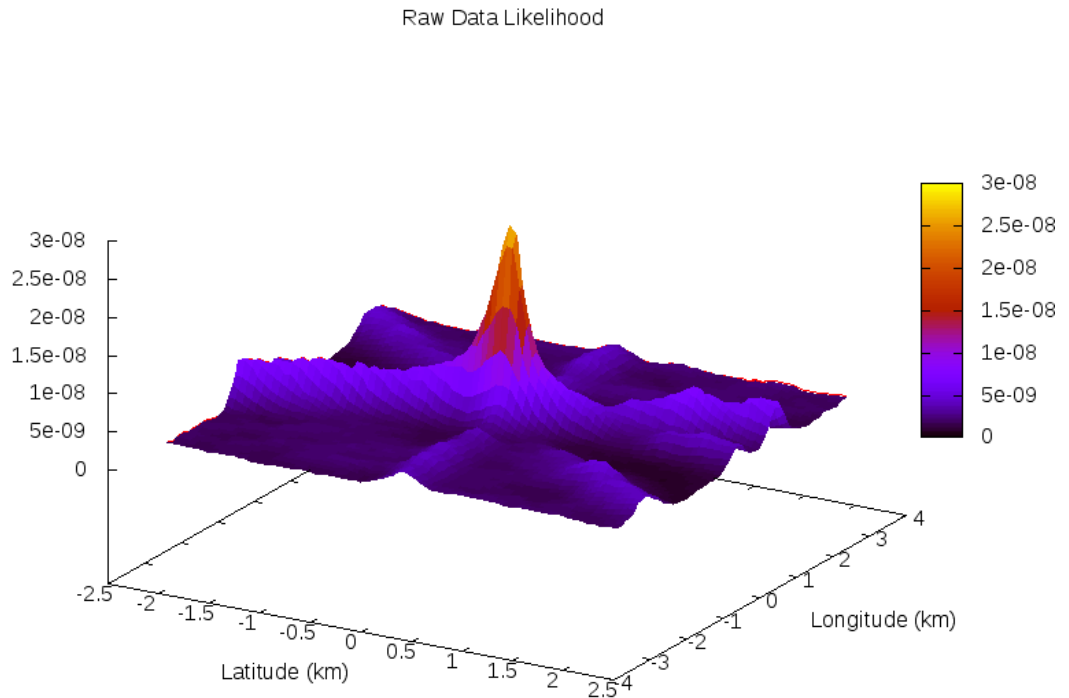


Figure 3.8: Raw data likelihood space, centered on known true receiver position, showing only latitude and longitude distributions conditional on all other parameters set to their known/estimated values. The raised ridges here can be thought of as wavefronts produced from each satellite.

In expressing the forward map mathematically many shortcuts were taken. For

example, the model of the GPS receiver is very basic. In reality many physical phenomena affect the signal inside the chip as it makes its way from antenna to digital output. All signal processing steps, such as that which occurs in the automatic gain control unit, should be modelled for a more accurate forward map. These all influence the probability density functions of the random variables used to model noise sources. The largest source of noise is likely the Gaussian thermal noise present at the receiver antenna, however, propagating a random variable through the receiver model analytically is quite difficult. Our likelihood function simply assumes an additive Gaussian noise added to the binary data at the output. The standard deviation σ of this noise model effectively scales the likelihood function exponentially, since

$$f(y_k|\mathbf{x}) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{k=1}^N [y_k - F_k(\mathbf{x})]^2\right) = \exp\left(-\frac{1}{2} \sum_{k=1}^N [y_k - F_k(\mathbf{x})]^2\right)^{\sigma^{-2}}. \quad (3.3)$$

Since the value of σ does not accurately correspond to any physical value, it is used instead to scale the magnitude of the likelihood in order to optimize MCMC performance. For example, Figure 3.8 shows the likelihood with $\sigma = 0.2$ which gives a nice plot visually, showing the more interesting structure. When it comes to exploring it with an MCMC however, a value of $\sigma = 0.05$ essentially wipes out the lower structure allowing the chain to focus on a more informative (roughly Gaussian) peak, as shown in Figure 3.9.

The Doppler and Phase steps also used this scaling technique but it was left out of their descriptions for clarity. Where the Doppler and Phase forward maps were both functions of 5 variables, we now have 39 to deal with. Of the 32 SV carrier phase offsets, however, the forward map is actually only dependent on those belonging to visible satellites, usually numbering between 5 and 14 at any given time from anywhere on Earth. Although the Doppler and Phase fix stages only loosely followed the Bayesian framework, they are critical to the success of this step. The marginal distributions found for each parameter of the Phase fix (approximated by Gaussians) are used now as prior probability density functions. The means of these prior pdfs are used to seed the MCMC, ensuring it starts off reasonably close to the mode. The prior's widths (as indicated by their standard deviations) keep the Markov chain from wandering too

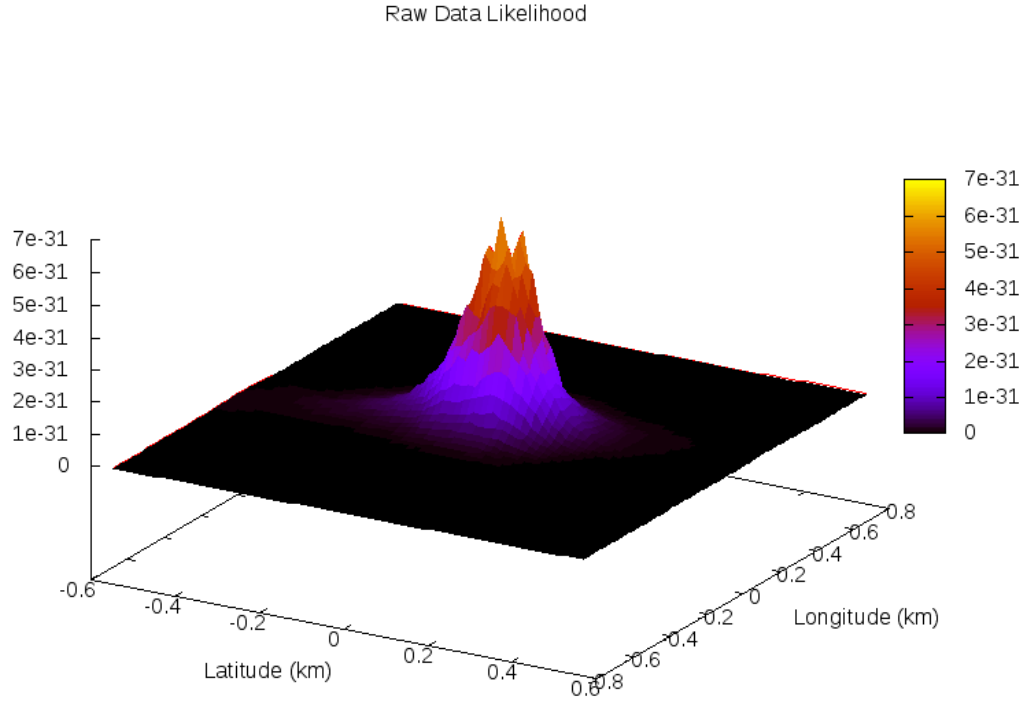


Figure 3.9: Raw data likelihood space, with standard deviation of noise set to $\sigma = 0.05$. By reducing σ the raised ridges of Figure 3.8 have been diminished considerably. This increases the effectiveness our simple MCMC design. The x and y axes are again relative to true position.

far from the mode and possibly getting trapped in a smaller mode elsewhere. Figure 3.10 shows a typical result of the Raw-data MCMC. The fix given by the means of the latitude and longitude in this example $(-45.8640, 170.5136)$ is now only 28.4 meters away from the true position. The standard deviations of the latitude and longitude are 78.3 meters and 102.2 meters respectively. The GPS time is now known to within roughly 0.3s and the error in altitude has also dropped considerably.

The three fixing stages were written in C++ and compiled into three executables. Scripts written in Ruby were used to automate the entire fixing process for a batch of stored data files. Each fix produced an output file in the JSON format which stored the

means and standard deviations of the states of each parameter in the Markov chain.

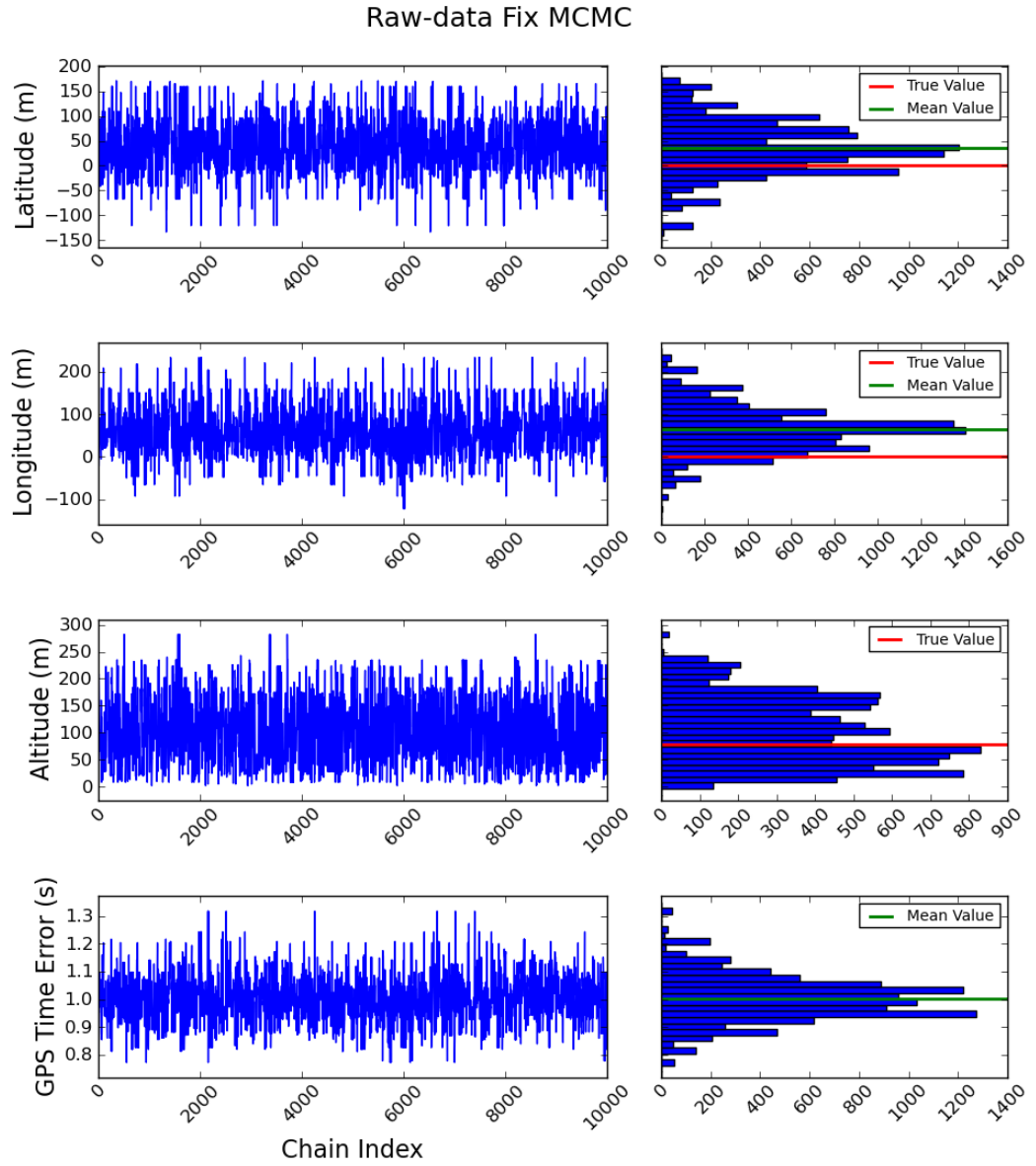


Figure 3.10: Running the Markov chain on the Raw-data likelihood shows our final posterior distributions. A position estimate based on the mean values is seen to be about 50 off in latitude and 100 meters off in longitude.

Chapter 4

Results

4.1 Doppler Fix Results

From the 12th to the 21st of May 2010, a GPS receiver was left on the roof of the Science III building at the University of Otago. Roughly every hour, the receiver captured and stored 4ms of binary data, sampled at 8.184MHz. In total 220 data files were created. The latitude and longitude means for each Doppler fix (as taken from the JSON fix files) were made into ‘Placemarks’ in a kml file then plotted on a map of the Earth using the Marble Desktop Globe software package (Version 0.9.5). They are shown in Figure 4.1, where the majority are seen to lie within a 100km radius of Dunedin city.

Using the known true position of the receiver, a latitude and longitude error was calculated for each fix. These are plotted in a histogram in Figure 4.2 along with a histogram of the latitude and longitude standard deviations (also found in the JSON fix files). We see the latitude and longitude standard deviations are largely concentrated around 25km and 70km respectively. The absolute position errors (excluding altitude), ie. distances from estimated lat/long to true lat/long for each fix, have a mean of 40.2km and a standard deviation of 65.6km.

To ensure that the Doppler fix standard deviations relate to position uncertainty, they are plotted against absolute error in Figure 4.3. The figure shows a loose linear correlation, ie. a higher standard deviation does indicate higher uncertainty of location and thus a less accurate estimate. It is reassuring to see there are no fixes with absolute

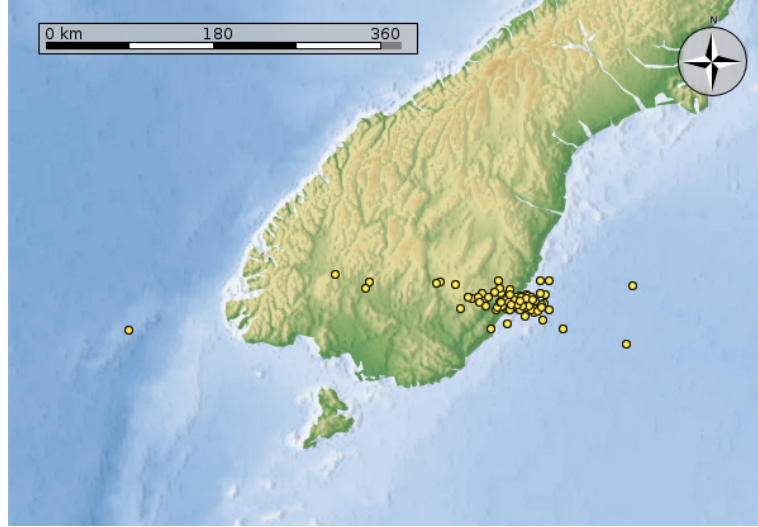


Figure 4.1: Doppler fix positions from 220 fixes. The majority of position estimates are congregated around Dunedin city.

errors much larger than their standard deviations, which would indicate either the Markov chain had been confined to an incorrect peak or that the forward map poorly simulated the system. The topmost red dot in Figure 4.3 shows the longitude fix with the highest error. This Doppler fix does lead to a wildly incorrect Phase fix ($\sim 900\text{km}$ away) and should ideally be excluded from further processing.

Based on the 300km separations between ridges in the Phase likelihood (as shown in Figure 1.3), a reasonable cutoff for longitude standard deviations in the Doppler stage may be 200km . Using this value to remove ‘bad’ Doppler fixes, 212 out of the 220 data files continue on to the Phase fix stage.

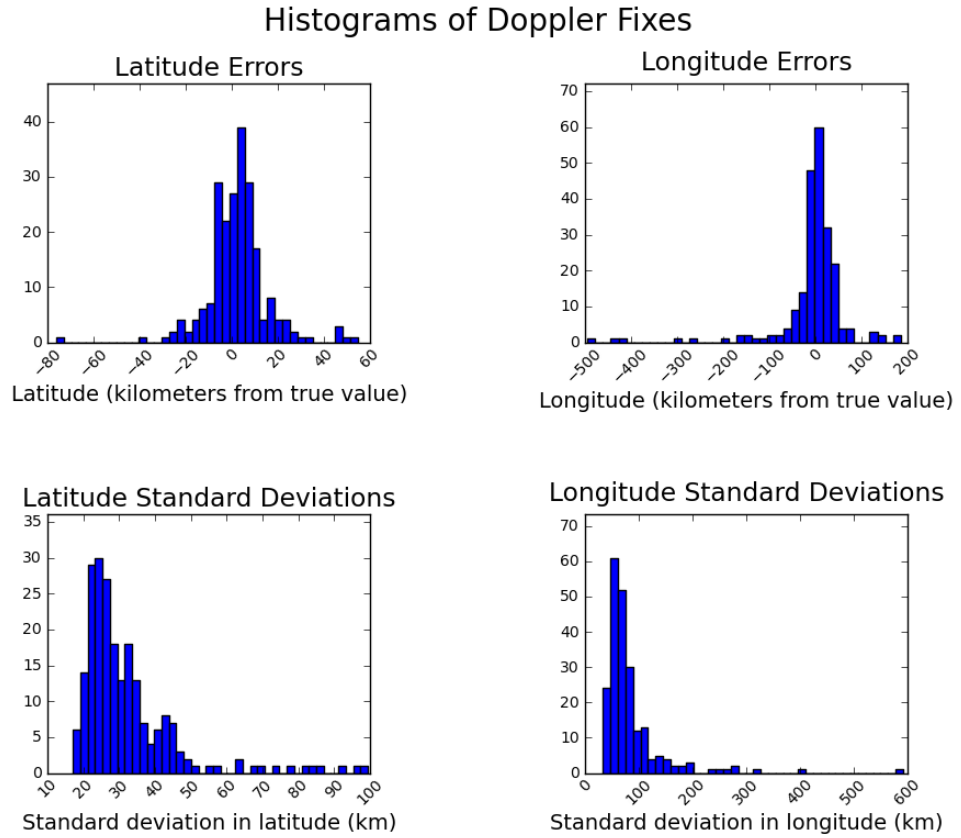


Figure 4.2: Top two histograms show the error magnitudes for the latitude and longitude position estimates for each of the 220 fix files. Bottom two histograms show the latitude and longitude standard deviations for each fix.

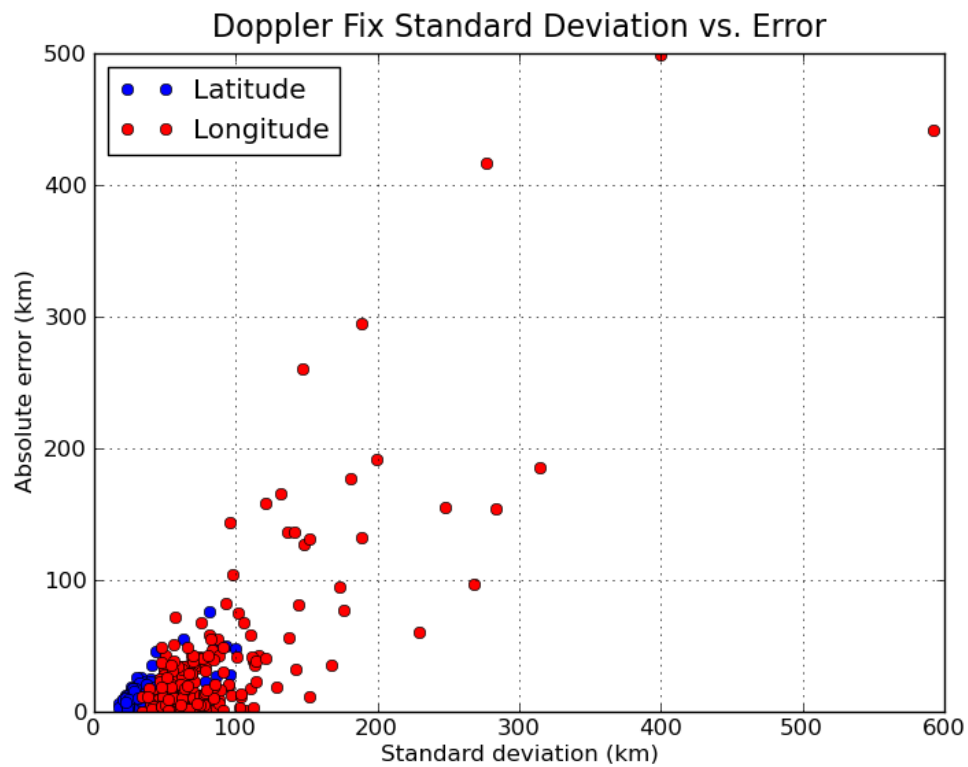


Figure 4.3: Plotting the standard deviations of each fix (for only the latitude and longitude parameters) againsts their absolute errors shows a roughly linear relationship. Fixes with higher standard deviations are more likely to give less accurate position estimates.

4.2 Phase Fix Results

The Phase fix results from the 212 data files that met the criteria for an acceptable Doppler fix are overlaid on a map of Dunedin, shown in Figure 4.4. Here we see they all lie within the city, with the worst fix (upper right corner) being only 7.7km out.



Figure 4.4: Phase fix positions from the 212 accepted data files overlaid on a map of Dunedin city. With the exception of one major outlier in the top right corner of the image, most fixes are in the city center (where the University of Otago campus and thus the GPS receiver are located).

Again the latitude and longitude standard deviations are plotted against absolute error in Figure 4.5, showing to a lesser degree the linear correlation seen for the Doppler fixes. Based on this figure, a suitable threshold for Phase fix latitude or longitude standard deviations could be around 1000m. Implementing this as an acceptance criterion leaves 205 of the 212 Phase fixes to be passed to the Raw-data fix stage. The absolute position errors (again excluding altitude) for these 205 ‘good’ fixes have a mean of 131.9m and standard deviation 134.8m, an improvement upon the Doppler fixes by a factor of over 300. Though choosing suitable cut-off criteria at each stage may seem

critical, it is really only a matter of convenience for a specific application. Should a particularly bad fix happen to meet the chosen criteria, it will most likely produce a fix in the following stage with a relatively high standard deviation, at which point it can be dealt with accordingly.

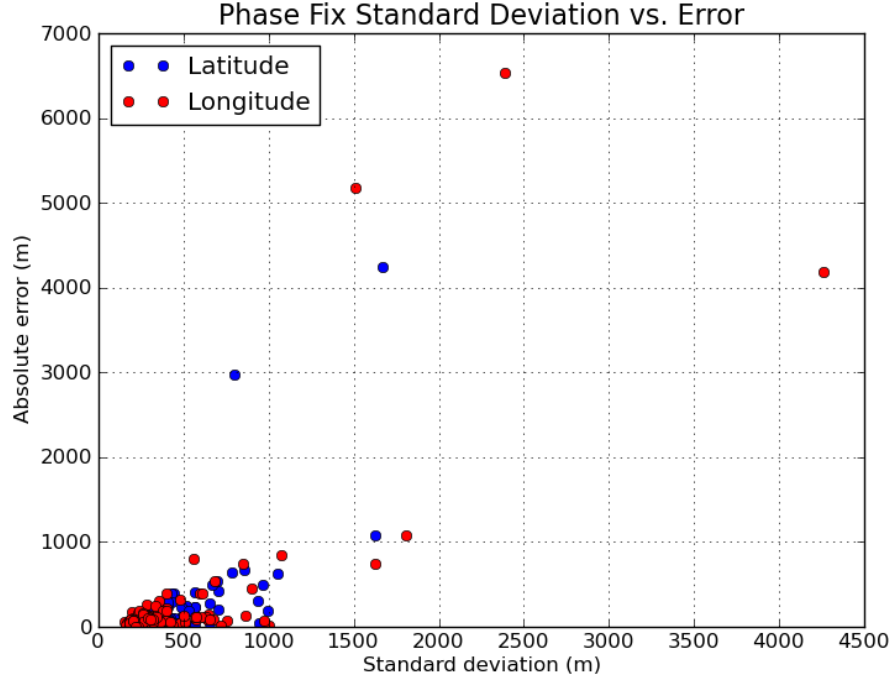


Figure 4.5: Plotting the Phase fix parameter standard deviations vs. errors for latitude and longitude shows that the least accurate fixes (those with higher absolute error) can be identified by their high standard deviations.

4.3 Raw-data Fix Results

4.3.1 Data requirements for the Raw-data fix

We have now reached the final stage in the fixing process. When evaluating the Doppler and Phase fixes, 4ms of data (sampled at $8.184\text{MHz} = 32736$ bits) was found to be a suitable amount to provide reliable results. It is interesting to note that the Raw-data

fix stage requires much less data to obtain a fix. To test this, a Raw-data fix was run on ten data files using 2ms of data (16368 bits) then rerun on the same files using only 0.4ms (3274 bits). The absolute error and standard deviation of the latitude and longitude parameters for each fix in the two cases is presented in Table 4.1

Table 4.1: Comparison of absolute error and standard deviation in latitude and longitude parameters for 10 fix files processed using 2ms and 0.4ms of data.

Date	Latitude				Longitude			
	Error (m)		Std. Dev.		Error (m)		Std. Dev.	
	2ms	0.4ms	2ms	0.4ms	2ms	0.4ms	2ms	0.4ms
2010-05-21 14:22:17	27.7	14.7	70.7	52.8	55.3	40.4	46.1	41.7
2010-05-21 15:22:14	42.1	71.9	69.6	267.9	4.4	147.9	65.2	309.7
2010-05-21 16:22:13	3.4	10.2	50.5	52.3	24.2	32.1	37.0	37.2
2010-05-21 17:22:17	8.0	8.1	50.7	52.5	66.0	90.8	64.0	48.2
2010-05-21 18:22:17	2.1	22.0	42.7	39.8	33.1	3.4	52.0	41.1
2010-05-21 19:22:14	17.0	35.6	73.3	65.4	83.9	20.4	119.7	75.1
2010-05-21 20:22:14	10.3	47.3	50.0	44.7	46.8	80.3	54.7	61.2
2010-05-21 21:22:17	127.1	111.8	115.0	161.2	164.8	131.5	114.8	183.5
2010-05-21 22:22:14	25.0	1.7	57.3	57.2	14.1	80.8	88.0	70.1
2010-05-21 23:22:14	46.7	9.6	153.4	64.4	13.0	49.9	52.7	37.9
Average:	30.96	33.28	73.32	85.81	50.54	67.77	69.43	90.57

An absolute position error ($\sqrt{lat_error^2 + long_error^2}$) was also calculated for each fix. Running Student's paired t -test on these two sets showed no significant difference (t -value = 0.9184, $df = 8$, $p = 0.3853$) in the errors obtained.

4.3.2 Raw-data fix results using 0.5ms

For the results presented in this section, 0.5ms of data was used in the Raw-data fix stage. Figure 4.6 compares the standard deviations and errors of the Raw-data fixes obtained from the 205 data files that made it through the Phase fix stage. Although there still exists a general trend of fixes with higher absolute errors having higher standard deviations, it is not as pronounced as in the Doppler fix case. This is likely

due to the Raw-data posterior distribution being more complex than the smooth, uni-modal Doppler posterior distribution. Although the Raw-data likelihood may look smooth and uni-modal in Figure 3.9, only the conditional probability distribution of two out of 39 dimensions is shown. The marginal distribution of any or even all of the other dimensions may include multiple peaks around the true value on which a Markov chain is easily trapped. This can lead to mean estimates representing the positions of incorrect peaks with relatively low standard deviations.

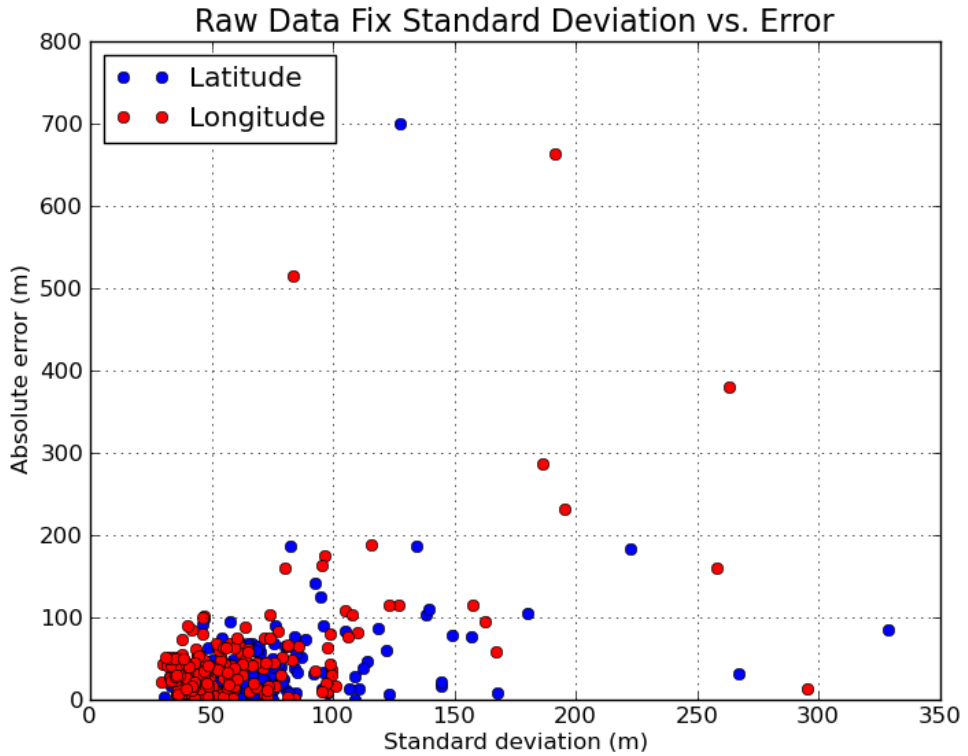


Figure 4.6: Raw-data fix parameter standard deviations vs. absolute errors. This closely resembles Figure 4.5 for the Phase fixes yet here there exist fixes where absolute error is as much as five times greater than standard deviation. These probably represent cases where the Markov chain has sampled an incorrect mode of the posterior distribution.

Cutting out all fixes with a latitude or longitude standard deviation greater than

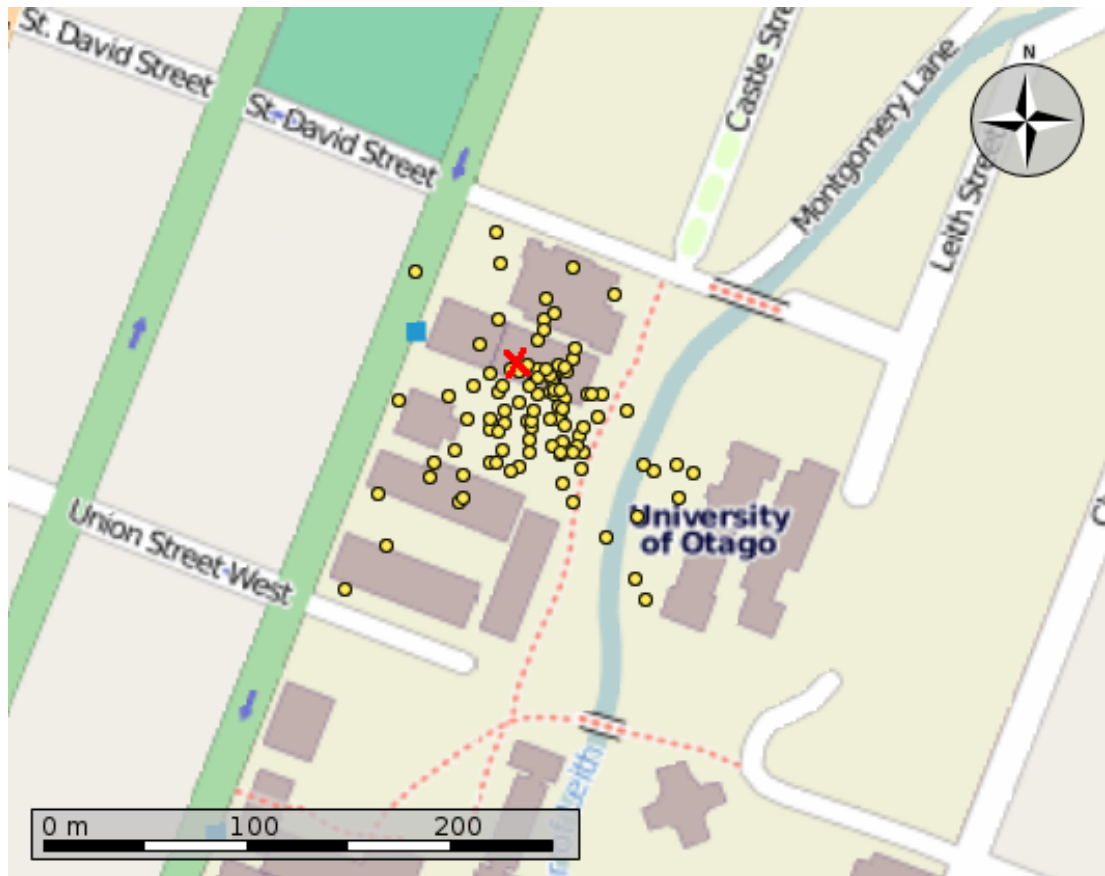


Figure 4.7: Final fix positions from the 185 accepted data files. All fixes lie within the University of Otago campus and are clustered around the Science III building (indicated with a red cross), the location of the GPS receiver.

120m, 185 out of the original 220 remain (84% ‘successful’) and are shown overlaid on a map of the University of Otago campus in Figure 4.7. The position estimates for the remaining 185 are all seen to lie within campus, where a red cross marks the true receiver position. The absolute position errors for these 185 fixes have a mean of 50.8m and standard deviation 29.7m.

Chapter 5

Discussion

Of the 220 data files used in testing the Bayesian inference approach, 185 returned a position estimate less than 250 meters from the receiver location. Moreover, every fix came with a measure of uncertainty for each parameter, information of great value for any scientific endeavor. Although there certainly remains room for improvement, these were the key objectives of this thesis.

5.0.1 Future Work

At the end of Section 4.3, the Raw-data fix was shown to require much less signal data than the Doppler and Phase fix stages, in order to obtain a position estimate. Thus, if we could engineer smarter ways to do the data reduction, the amount of data required for a successful fix could potentially be reduced by a factor of ten. Acquiring a GPS fix from only a few tenths of a millisecond worth of signal data would indeed be an improvement on the often minute long wait required by many standard GPS receivers.

One of the key benefits of the Bayesian inference approach is the inherent ability to take prior information. This is of particular interest in a practical application such as animal tracking. If a tag were to capture signal data every 5 minutes say, one can estimate the maximum distance the particular creature could travel in such a time. By combining this with the position information of the previous fix, a prior probability density can incorporate such knowledge. The prior information could then be sent straight to the Raw-data fix stage, eliminating roughly two thirds of the processing

time required. Furthermore, if this method proved effective, the amount of signal data captured by the receiver could be reduced to the mere 0.4ms required by the Raw-data fix, with only every tenth (or more) fix capturing the full 4ms. This would further reduce the power demands of the receiving chip.

5.0.2 Limitations

One of the critical limitations encountered in the project was the use of a generic estimator applied to the Markov chain histograms. Using the mean as an estimator assumes a symmetrical and uni-modal distribution. Although many of the fixes' posterior distributions fit well to this approximation, it was likely the cause of the greatest outliers. Many, if not all, of the discarded data files still contain enough information to produce a successful fix. For example, visually inspecting the Markov chain histograms of a 'bad' Doppler fix, one may have a more intuitive sense of a good position estimate rather than simply calculating the mean value.

Another obvious limiting factor is the simplification of the (Raw-data) forward map used to model the physical system. No absolute measures of signal amplitude were included, despite the output power at a satellites antenna and the thermal noise power spectral density at the receiver antenna being well known. The effects of the Earth's ionosphere and a large portion of receiver electronics were also ignored. By including such factors we could expect the results of the Raw-data fixes to improve closer toward those achieved in standard GPS receivers.

It is worth touching again upon the importance of the σ parameters found in each of the three likelihood functions. As outlined in Section 3.3, these values are adjusted by the user in order to scale the likelihoods. As a consequence, the value of σ also scales the standard deviations of our posterior distribution histograms. Although the posterior standard deviations are not a direct measure of the true uncertainty of one's position, they still hold value within a certain scope of situations. For example, in comparing fixes obtained from the same receiver (where σ can remain constant), the standard deviations can be used as a relative measure of uncertainty. Keeping a record of typical values obtained from a specific GPS receiver, one can get a feel for the validity

of a fix based on these values.

5.0.3 Conclusion

Using Bayesian inference techniques to get the most information from limited data has proven to be a very successful approach to GPS position estimation. Signal processing methods often provide quick and easy solutions to problems like GPS positioning. In situations where processing power is readily available, Bayesian analysis of data can usually give more in depth results. The future work discussed introduces ample room for improving the results obtained and the methods employed can be easily adapted to suit any number of similar applications.

Appendix

WGS84 to ECEF Coordinate Transformation

In order to calculate a satellite's signal doppler shift for a given receiver position and time, we need to find the velocity vector of the SV in earth centered, earth fixed (ECEF) coordinates. The broadcast ephemerides (which are coded into the each SV's signal or available online) define parameters of each SV's position and motion with respect to their plane of Keplerian orbit. These parameters are listed in Table 1. To transfer from this Keplerian coordinate system to ECEF we refer to the GPS Interface Control Document [3].

Firstly, we need to make a few adjustments and corrections to the given parameters in order to calculate the SV's position for an arbitrary time t .

Using the new parameters defined in Table 2, the SV's position vector in the orbital plane is given by

$$\mathbf{r}_{icd} = \begin{bmatrix} r_k \cos u_k \\ r_k \sin u_k \\ 0 \end{bmatrix} = \begin{bmatrix} x_{icd} \\ y_{icd} \\ z_{icd} \end{bmatrix} \quad (1)$$

where 'icd' signifies the orbital plane coordinate system as used in the GPS ICD documentaion. The corresponding earth centered, earth fixed coordinates (ECEF) are then found by a rotation by the angle of inclination followed by a rotation by the angle of the longitude of ascending node. The two rotation matrices are [4]

$$R_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad R_3(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Table 1: Broadcast ephemeris parameters describing a satellite's motion.

M_0	Mean anomaly at reference time
Δn	Mean motion difference from computed value
e	Eccentricity
$(A)^{1/2}$	Square root of the semi-major axis
$(OMEGA)_0$	Longitude of ascending node of orbit plane at weekly epoch
i_0	Inclination angle at reference time
ω	Argument of perigree
$OMEGADOT$	Rate of right ascension
$IDOT$	Rate of inclination angle
C_{uc}	Cosine harmonic correction term to the argument of latitude
C_{us}	Sine harmonic correction term to the argument of latitude
C_{rc}	Cosine harmonic correction term to the orbit radius
C_{rs}	Sine harmonic correction term to the orbit radius
C_{ic}	Cosine harmonic correction term to the angle of inclination
C_{is}	Sine harmonic correction term to the angle of inclination
t_{oe}	Reference time ephermis
$IODE$	Issue of data (ephemeris)

and the SV's postion is then given by

$$\mathbf{r} = R_3(-\Omega_c) \cdot R_1(-I) \cdot \mathbf{r}_{icd} \equiv R_i^e \cdot \mathbf{r}_{icd} \quad (3)$$

where R_i^e is the rotation matrix from the ICD orbit to ECEF system and has the following form

$$R_i^e = \begin{bmatrix} \cos \Omega_k & -\sin \Omega_k \cos i_k & \sin \Omega_k \sin i_k \\ \sin \Omega_k & \cos \Omega_k \cos i_k & -\cos \Omega_k \sin i_k \\ 0 & \sin i_k & \cos i_k \end{bmatrix} \quad (4)$$

thus $\mathbf{r}$ can be found from

$$\mathbf{r} = \begin{bmatrix} x_{icd} \cos \Omega_k - y_{icd} \cos i_k \sin \Omega_k \\ x_{icd} \sin \Omega_k - y_{icd} \cos i_k \cos \Omega_k \\ y_{icd} \sin i_k \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (5)$$

Table 2: Paramaters defining the WGS84 coordinate system.

$\mu = 3.986005e14$ $\dot{\Omega}_e = 7.2921151467e - 5$ $A = (\sqrt{A})^2$ $n_0 = \sqrt{\mu/A^3}$ $t = t - t_{oe}$ $n = n_0 + \Delta n$ $M_k = M_0 + nt_k$ $E_k = M_k + e \sin E_k$ $\nu_k = \tan^{-1} \frac{\sqrt{1-e^2} \sin E_k}{\cos(E_k - e)}$ $\Phi_k = \nu_k + \omega$	Earth's universal gravitational parameter Earth's rotation rate Semi-major axis Computed mean motion (rad/sec) Time from ephemeris reference epoch Corrected mean motion Corrected mean anomaly Kepler's equation eccentric anomaly True anomaly Argument of latitude
Second Harmonic Pertubations	
$\delta u_k = c_{us} \sin 2\Phi_k + c_{uc} \cos 2\Phi_k$ $\delta r_k = c_{rs} \sin 2\Phi_k + c_{rc} \cos 2\Phi_k$ $\delta i_k = c_{is} \sin 2\Phi_k + c_{ic} \cos 2\Phi_k$	Argument of latitude correction Radius correction Inclination correction
$u_k = \Phi_k + \delta r_k$ $r_k = A(1 - e \cos E_k) + \delta r_k$ $i_k = i_0 + \delta i_k + (IDOT)t_k$ $\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e)t_k - \dot{\Omega}_e t_{oe}$	Corrected argument of latitude Corrected radius Corrected inclination Corrected longitude of ascending node

Finding the satellites velocity in the ECEF system can now be found by differentiating 5, [8]

$$\dot{\mathbf{r}} = \dot{R}_i^e \cdot \mathbf{r}_{icd} + R_i^e \cdot \dot{\mathbf{r}}_{icd} \quad (6)$$

, or by simply calculating $\mathbf{r}$ for one second ahead and one second behind the desired time then dividing the difference by two.

$$\mathbf{v} \approx \frac{\mathbf{r}(\mathbf{t} + \mathbf{1}) - \mathbf{r}(\mathbf{t} - \mathbf{1})}{2} \quad (7)$$

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